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Wind power forecasting system with data enhancement and algorithm improvement

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ABSTRACT

Wind power generation has strong volatility. Accurate wind speed forecasting can not only avoid the waste of power resources, but also facilitate the development of clean energy and promote the energy transition worldwide. However, previous research has predominantly focused on the accuracy of wind power prediction, while ignoring the reliability of wind speed prediction system. In this research, a hybrid forecasting system with both accuracy and reliability of wind power forecasting is proposed. Firstly, a hybrid adaptive decomposition denoising algorithm is proposed to solve the unreasonable decomposition and residual noise. To improve the search performance, the seagull algorithm is optimized by chaotic system and Cauchy operator, and then the parameters of long short-term memory model are adjusted. Finally, based on data enhancement theory, an interval prediction model combined with kernel density estimation is proposed. The model is verified by the historical data of Sotavento wind farm in Spain and Eman wind farm in China. The average absolute percentage error values of wind speed point prediction are 2.87% and 8.01%, respectively. At the same confidence level, the interval prediction model proposed has narrower widths compared to the comparative model, with higher average interval scores. The results indicate that the point prediction model proposed in this research exhibits higher accuracy, while the interval prediction model demonstrates greater stability and reliability. These findings provide technical support for wind power forecasting.

1. Introduction

The greenhouse effect has been one of the key issues of global concern. It arises due to the emission of greenhouse gases produced by fossil energy. To further reduce greenhouse gas emissions and better meet the challenges posed by climate change, promoting the development of renewable energy is a necessary means [1]. According to the latest report of the International Renewable Energy Agency, to achieve a successful energy transition and control the global temperature risk, the installed capacity of renewable energy needs to increase from about 3000 GW (GW) now to more than 10,000 GW in 2030, so as to promote the energy transition dominated by renewable energy [2].

Wind energy, as one of the renewable energy sources, is widely developed and utilized all over the world, of which wind power is again the most popular application of wind energy. Advance forecasting of wind energy is conducive to grid system integration, improving power system consumption problems, and rationally utilizing wind energy to reduce greenhouse gas emissions. Therefore, it is necessary to forecast wind energy. However, the renewable energy generation represented by wind power has the characteristics of strong volatility and fast real-time change. The randomness and intermittency of wind energy leads to frequent wind abandonment, which seriously affects the utilization of wind energy. Both the volatility and randomness of wind power can lead to inaccurate wind power prediction, which affects the safe and stable power supply of the grid system and causes the waste of wind energy.

To help grid scheduling, researchers have conducted a lot of studies on the determinism and uncertainty of wind power. Ye et al. [3] proposed a comprehensive method for short-term wind power prediction based on frequency analysis, fluctuation clustering and history matching to improve the accuracy of wind power prediction. Che et al. [4] established a prediction model with quantitative uncertainty to guarantee grid scheduling and operation. Wang et al. [5] developed a prediction model for wind power prediction through the Integral fractional quantile regression for interval prediction of wind power sequences to quantify the uncertainty of wind speed fluctuations. Mansoor et al. [6],

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Abbreviation

AIS	Average interval score
Bi-GRU	Bidirectional gated recurrent unit
BP	Back Propagation
CEEMDAN	Complete ensemble empirical Mode decomposition with adaptive noise
CWC	Coverage width-based criterion
EEMD	Ensemble empirical mode decomposition
EMD	Empirical mode decomposition
FA	Factor analysis
GA	Genetic optimization algorithm
GCDLA	Graph convolutional deep learning architecture
GRU	Gated Recurrent Unit
GW	Gigawatt
HADD	Hybrid adaptive decomposition denoising
ICA	Independent component analysis
IMF	Intrinsic Mode Functions
IRENA	International Renewable Energy Agency
ISOA	Inertia seagull optimization algorithm
KDE	Kernel density estimation
KPCA	Kernel principal component analysis
KS	Kolmogorov-Smirnov
LDA	Linear discriminant analysis
LLE	Local linear embedding
LMD	Local mean decomposition
LSTM	Long short-term memory
MAE	Mean absolute error
MAPE	Mean absolute percentage error
MODWT	Maximum overlap discrete wavelet transform
NWP	Numerical weather prediction
OOA	Osprey optimization algorithm
PCA	Principal component analysis
PICP	PI coverage probability
PINAW	PI normalized averaged width
PSO	Particle swarm optimization
RMSE	Root mean square error
RNN	Recurrent neural network
$\mathbf{R^2}$	Coefficient of determination
SARIMA	Seasonal autoregressive integrated moving average
SMOTE	Synthetic minority over-sampling technique
SOA	Seagull optimization algorithm
SOM	Self-organizing mapping
SSD	Singular spectrum decomposition
TA	Time attention
VMD	Variational mode decomposition
WD	Wavelet decomposition
WOA	Whale optimization algorithm
WST	Wavelet soft threshold
WT	Wavelet transform

to address the unsatisfactory performance of the existing prediction models in short-term power prediction, proposed a new hybrid model based on the intelligent model of hybrid stochastic algorithms, and at the same time, enhanced the convergence through accelerated computation, thus improving the degree of accuracy of the prediction. Jin et al. [7], for the inherent intermittency and stochasticity of wind energy that leads to the degradation of the performance of the prediction model, an adaptive method based on the integration of offline global and online local learning options is established, thus improving the accuracy and reliability of wind prediction. Fan et al. [8] introduced fluctuation pattern recognition to quantify wind speed volatility by considering the probabilistic prediction of wind power under the fluctuation pattern of wind power. Farah et al. [9] performed wind power prediction in multiple steps as well as at different elevations and developed a prediction model with high prediction accuracy and faster learning speed on long series. Therefore, Improving the accuracy of wind speed prediction and quantifying the risk of wind speed fluctuation are still the focus and difficulty of current research.

Wind speed prediction accuracy is closely related to original data features, prediction model selection and error correction methods. The research directions at home and abroad can be divided into the following three aspects: (1) data processing algorithm based on data characteristics; (2) prediction model algorithm aiming at accurate prediction; (3) interval prediction method aiming at quantifying data fluctuation.

Data processing methods are mainly divided into two parts: data screening and data decomposition. Data filtering algorithm can improve data quality by cleaning invalid data in original data set. Through data dimensionality reduction, some features with the greatest impact on the dependent variable are found to participate in the subsequent modeling process [10]. Reference [11] adopted the maximum overlap discrete wavelet transform (MODWT) for noise removal. Then the random forest algorithm is used to calculate the importance of variables, reduce the dimension, and reduce the number of variables involved in modeling. In Ref. [12], independent component analysis (ICA) was introduced to improve the variable screening effect. Combined with the adaptive decomposition method, the boundary effect problem in the process of decomposition was effectively solved. Zhang et al. combined principal component analysis (PCA) with neural network to eliminate redundant information and improve the accuracy of the prediction model [13]. In addition, common data screening methods include linear discriminant analysis (LDA) [14], factor analysis (FA) [15], local linear embedding (LLE) [16] and kernel principal component analysis (KPCA) [17].

The data decomposition algorithm is to decompose complex sequences into multiple relatively simple components to further reduce the difficulty of analysis modeling. Commonly decomposition algorithms are used for data feature extraction include singular spectrum decomposition (SSD) [18], wavelet decomposition (WD) [19] and empirical mode decomposition (EMD) [20]. Paper [21] proposed an Ensemble Empirical Mode Decomposition (EEMD) algorithm and feature selection methods to extract important variables and further improve the accuracy of predictions. Tian et al. implemented wind speed data decomposition and feature extraction using EMD and local mean decomposition (LMD) algorithms, and selects different models for prediction based on the different characteristics of odd and even sequences [22]. To solve the over-decomposition problem of VMD, paper [23] introduced energy entropy theory to determine the number of decomposition modes and combines sample entropy (SE) theory to determine the complexity of different components. Zhang et al. proposed a fully integrated empirical modal decomposition algorithm based on wavelet transform (WT) and adaptive noise reduction (CEEMDAN) to effectively remove noise during the original data acquisition process [24].

However, the traditional signal processing method has many defects. For example, WD needs to determine the wavelet basis function in advance [25], mode aliasing exists in EMD [26], and the selection of the number of VMD decomposition needs to be preset [27]. CEEMDAN, as an improved algorithm of EMD, can effectively solve mode aliasing and end effects [28]. It has high computational efficiency and good robustness. This research proposes a hybrid adaptive decomposition denoising algorithm. CEEMDAN was used for data decomposition, combined with self-organizing mapping (SOM) network and wavelet soft threshold (WST) method for signal denoising [29], so as to reduce the influence of noise on prediction results.

Wind power forecasting models are mainly divided into physical

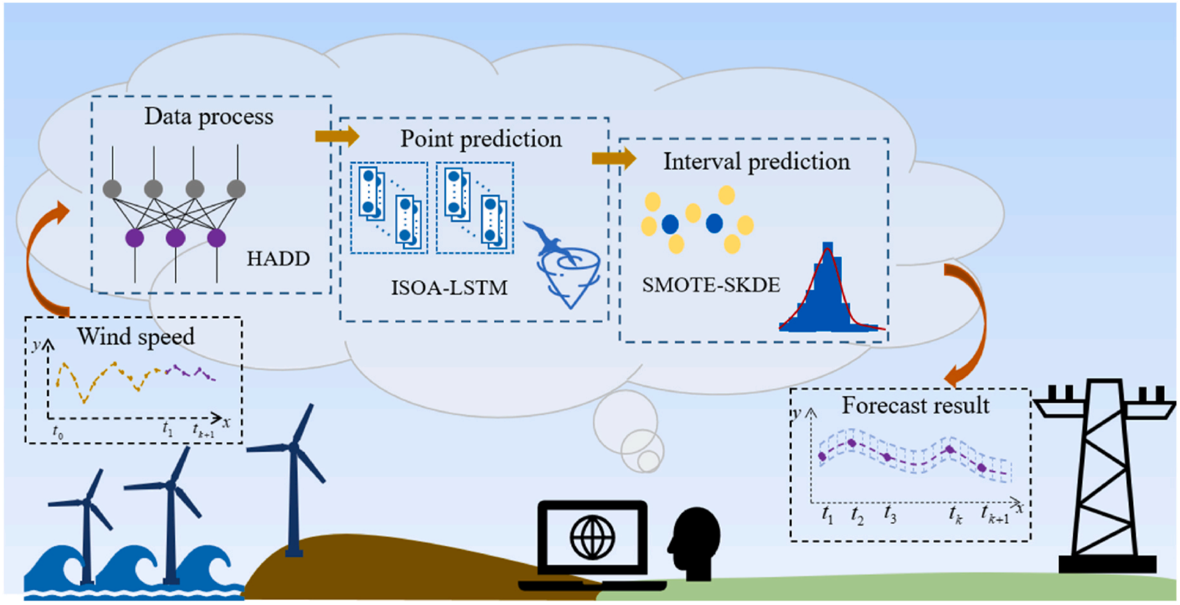


Fig. 1. Schematic diagram of the Hybrid wind power forecasting system.

model, statistical model and neural network model. Numerical weather prediction (NWP) model is the most common physical model for wind speed prediction [30], and accurate wind speed prediction can be achieved through detailed observation records of atmospheric conditions and environmental variables. Traditional statistical methods are mainly used to predict stationary time series. The Seasonal Autoregressive Integrated Moving Average (SARIMA) model is used to predict wind speeds for offshore and offshore wind farms in Scotland. Compared with the Gated Recurrent Unit (GRU) and Long Short-Term Memory (LSTM) algorithms based on deep learning, SARIMA is more robust and accurate [31]. Although on some data sets, statistical methods show good predictive effect, for complex and variable wind speed data, the neural network model with strong nonlinear processing ability can achieve more robust prediction. By combining the traditional Autoregressive Integrated Moving Average (ARIMA) model with the BP model, the correlation between wind and wave is analyzed, and the historical wind speed data is fully utilized to further improve the accuracy of the wave prediction model [32]. In recent years, the deep learning model has been developed rapidly, showing good performance in extracting the inherent rules of data and realizing complex classification prediction [33]. In work [34], deep learning model was used to predict wind speed, feature extraction was carried out by LSTM, and wind speed at different locations was predicted by combining graph convolutional deep learning architecture (GCDLA). Research [35] used bidirectional gated recurrent unit (Bi-GRU) to forecast ultra-short-term wind power. The feature attention (FA) and time attention (TA) mechanisms were introduced to further improve the prediction effect of the model. The improper parameter initialization of neural network will lead to the prediction deviation. The use of optimization algorithm improves the prediction accuracy of the model. Common optimization algorithms include genetic optimization algorithm (GA) [36], particle swarm optimization algorithm (PSO) [37], whale optimization algorithm (WOA) [38] and osprey optimization algorithm (OOA) [39]. Seagull optimization algorithm (SOA) is a swarm intelligent optimization algorithm proposed in 2018 [40], which imitates the migration and aggressive behavior of gull and has strong search ability. In this research, the deep learning model LSTM model is selected to predict wind speed data and achieve high precision prediction. The optimization search of seagull is improved using chaotic mapping and Cauchy distribution theory. Then, improved seagull optimization algorithm (ISOA) is used to optimize the parameters of the LSTM model to improve the accuracy of the high point prediction model.

The interval prediction method quantifies the fluctuation range of data by analyzing the error sequence. In studies [41], Zhang et al. proposed an uncertain prediction model based on Monte Carlo method that can adapt to different error characteristics. Jiang et al. proposed a combined prediction model based on sample entropy and conditional kernel density estimation to complete interval prediction [42]. However, the error data has strong randomness [43], and the existing error correction models have poor generalization ability and low operation efficiency [44,45]. To improve the reliability of wind speed interval prediction model, data enhancement technology is introduced in this research. Put synthetic minority over-sampling technique (SMOTE) sampling on the error sequence [46], get the extended error sequence, then combine with the kernel density estimation method [47], establish the range of wind speed prediction. This method effectively improves the running efficiency of the model and avoids the overfitting problem caused by Bootstrap resampling.

There are still some problems in the research of wind power prediction: (1) the noise signal obtained by the decomposition algorithm is improperly processed, and the effective signal information is lost, which greatly reduces the subsequent prediction effect; (2) Traditional optimization algorithms are easy to fall into local optimum, weak search ability and slow convergence speed; (3) Due to the small amount of error data obtained in practical applications, it is easy to overfit when dealing with random error sequences during interval estimation. (4) The single wind speed prediction model has low accuracy, poor performance, and can not effectively quantify the uncertainty.

To solve those problems, this study proposes a new prediction model, which is innovative and practical. Firstly, this research proposes a new hybrid adaptive decomposition and denoising (HADD) algorithm, which can adaptively decompose and denoise the original wind speed data. Through this algorithm, can effectively eliminate the noise and interference in the data and improve the accuracy and stability of the prediction model. Secondly, this work utilize the ISOA algorithm to optimize the LSTM parameters. By continuously adjusting and optimizing the LSTM parameters, the model can be better adapted to the characteristics of the data, thus improving the prediction performance. According to the optimized model, preliminary prediction results is obtained. Finally, through the in-depth analysis of the error series, propose a kernel density wind speed interval prediction model based on data augmentation techniques. This model can quantify the fluctuation range of wind speed, which provides a new idea and method for the field of wind speed prediction. The Schematic diagram of the Hybrid wind power forecasting system is shown in Fig. 1. The novelties and contributions of this study are as follows:

(1) A new hybrid adaptive decomposition and denoising (HADD) algorithm is proposed. By combining CEEMDAN, SOM and WST algorithms, the proposed algorithm can effectively avoid pattern aliasing, reduce residual noise and improve data quality.
(2) A hybrid improvement strategy is adopted to improve the efficiency of the genetic Optimization algorithm (SOA). Firstly, by introducing chaotic mapping, the population diversity of the algorithm was increased, so as to improve the local search ability of the algorithm and avoid falling into the local optimal solution prematurely. Secondly, to further improve the convergence ability, the cosine function is introduced to control the moving direction of the position of the seagulls. The cosine function has the characteristics of periodicity and volatility, which can make the seagulls gradually converge to the vicinity of the optimal solution during the search process. By adjusting the parameters of the cosine function, the ability of exploration and exploitation can be balanced, so as to better find the global optimal solution. In addition, to avoid the optimization falling into the local optimal solution, Cauchy mutation perturbation operator is used to improve the global search ability.
(3) A kernel density interval estimation model based on data augmentation algorithm is proposed. The random error sequence is first processed using the Smote algorithm to avoid the overfitting problem. The Smote algorithm balances the class distribution in the dataset by generating synthetic samples, thereby reducing the impact of random errors. The model combines the kernel density estimation method to solve the problem of unknown error distribution. Kernel density estimation is a non-parametric estimation method, which can estimate the density function of data from a given set of data points. By applying kernel density estimation to the processed data, a more accurate and reliable interval estimate can be obtained.
(4) A wind speed prediction system combining data decomposition, improved optimization algorithm, deep learning and interval estimation is established. Firstly, the data decomposition method was used to decompose the complex wind speed series into simpler sub-series, so as to better analyze the wind speed change law. Secondly, the optimization algorithm is improved to improve the accuracy and efficiency of prediction. Then, the neural network model is optimized by the optimization algorithm, which can deal with complex characteristics of wind speed components, so as to improve the accuracy of prediction. Finally, the interval estimation method was used to quantify the fluctuation range of wind speed, which could more accurately describe the fluctuation range of wind speed and improve the stability and reliability of the prediction system.

The sections of this research are arranged as follows. Chapter 2 introduce HADD, ISOA based on chaotic mapping and Cauchy variation, LSTM and kernel density interval prediction (SKDE) theory based on data enhancement. Chapter 3 introduces the steps of the model establishment and the error index of this work. In chapter 4, the deterministic prediction results on the model real wind farm data set are given. In chapter 5, the effect of the proposed interval prediction model is introduced. The sixth chapter is the summary of this research, and the future development direction to make a prospect.

2. Theoretical basis of methods

This chapter mainly introduces the methods used in the model and the improvement theory. Firstly, HADD is proposed to solve the noise existing in wind farm data acquisition and decomposition. Secondly, an improved Seagull algorithm (ISOA) based on mixed chaotic mapping and Cauchy variation is proposed, which improves the population diversity and global search ability of the optimization algorithm. Finally, on the basis of deterministic prediction, a data enhancement technique is applied to the error sequence and kernel density estimation is combined to construct wind speed interval prediction. The following is the specific content of the methodology.

2.1. Hybrid adaptive decomposition denoising algorithm (HADD)

Common decomposition-based denoising algorithms usually treat the component with the highest frequency as the noise signal and directly manipulate this component. This method can cause the deviation of the predicted results from the actual values. Therefore, based on the shortcomings of existing algorithms, this study proposes a hybrid adaptive decomposition denoising algorithm that combines CEEMDAN, SOM, and wavelet soft threshold denoising methods adaptively [48]. The decomposed signal components are obtained using CEEMDAN. The self-organizing mapping neural network is introduced to classify components [49], and the components whose classification results are noise signals are denoised to further reduce the residual white noise. This algorithm makes the prediction result more robust.

2.1.1. Complete Ensemble Empirical Mode Decomposition with adaptive noise

CEEMDAN proposed by Torres in 2011 can better solve the problem of modal aliasing, reduce noise residue, and improve computational efficiency, and has been widely used [50]. The advantage of CEEMDAN method lies in its ability to adaptively add appropriate white noise, effectively improving the signal-to-noise ratio [51]. It generates a new decomposition sequence based on each residue to minimize noise interference. As a result, the error between the reconstructed signal after decomposition and the original signal is extremely small, which accurately reflects the characteristics of the original signal [52].

$E_j(\cdot)$ is defined as the component obtained by EMD decomposition, $\varepsilon_i(i=1,2,\ldots)$ is defined as the signal-to-noise ratio of the $(i+1)-th$ time, and the specific steps of CEEMDAN decomposition are as follows:

(1) Add Gaussian white noise $N_1(t), N_2(t), \ldots, N_i(t), \ldots, N_k(t)$ with different signal-to-noise ratio with ε_0 to the original sequence $x(t)$, decompose the sequence by EMD method to obtain IMF_1^i, whose mean value is the first IMF_1, and the process is shown in the following equation.

$$IMF_1 = \frac{1}{k}\sum_{i=1}^{k} IMF_1^i = \frac{1}{k}\sum_{i=1}^{k} E_1(x(t) + \varepsilon_0 N_i(t)) \tag{1}$$

(2) Calculate the first order residual $e_1(t)$.
(3) The first-order residual $e_1(t)$ is used as the new sequence to be decomposed, and the Gaussian white noise with the signal-to-noise ratio ε_1 is added to the new sequence, which is expressed as: $x_{new1} = e_1(t) + \varepsilon_1 E_1\{N_i(t)\}$. After EMD decomposition, the second mode IMF_2 is obtained, and the process is expressed as follows:
(4) Calculate the $j-$th residual sequence.
(5) Compute the $(j+1)-th$ mode: Decompose sequence $e_j(t) + E_j(N_i(t))$ with EMD to obtain the IMF_{j+1}.
(6) Return to step 4 and continue the calculation until the final residual $e_N(t)$ satisfies the iteration termination condition or reaches the maximum number of iterations, and the iteration stops. The original signal $x(t)$ conforms to the formula:

$$x(t) = \sum_{i=1}^{j+1} IMF_i + e_N(t) \tag{2}$$

2.1.2. HADD algorithm implementation scheme

CEEMDAN does not need to set the number of modes, and the decomposition speed is fast and the effect is good. However, there will be noise residual due to the process of wind speed data acquisition and decomposition. Therefore, components are classified by SOM method, which is an unsupervised and self-learning clustering network [53]. It consists of the input layer and output layer and is linked by the weight vector [54]. The output neuron is connected to other nodes in the domain, and the nodes are activated by competing with each other. The component signals containing noise are identified by SOM and then processed with wavelet soft threshold denoising [55].

Wavelet denoising is an algorithm to effectively separate the wavelet transform of signal from the wavelet transform of noise in the wavelet domain. The effective signal has a certain continuity in the time domain, while the noise still has a strong randomness after wavelet transform, which can be removed according to the mean square error value [56]. The wavelet threshold denoising method consists of three steps: wavelet decomposition, threshold processing and signal reconstruction. The process of threshold determination has great influence on the denoising result. In the process of calculating the hard threshold function, there will be signal oscillation phenomenon and the denoising effect is poor. Therefore, using soft threshold function for denoising can avoid the generation of signal break and the signal smoothing effect is better [57].

Each operation process of HADD is as follows.

Step 1: White noise with different signal-to-noise ratios is added to the original wind speed data x(t), and the corresponding components $IMF_1, IMF_2, \ldots, IMF_k$ are obtained by CEEMDAN decomposition.
Step 2: Initialize the SOM network. The weight ω_j of each neuron in the output layer is assigned a random initial value in [0,1], and the learning rate η is randomly initiated, $0 < \eta < 1$.
Step 3: IMF_1 is set as the standard component sample, and the input component vector is normalized. Each node of the output layer is traversed. Here, Euclidean distance is selected as the discriminant rule. Calculate the value between the standard component sample and each output node, and select the minimum distance as the winning node.

$$d_j = \sqrt{\sum_{p=1}^{P} \left(x_p - w_{pj}\right)^2} \tag{3}$$

where, d_j is the distance between the weight vector of neurons in the output layer and the input vector. ω_{pj} is the weight of the j-th neuron in the output layer of the p-th input vector. x_p is the value of the input vector.

Step 4: Update the weights in the domain of winning nodes and adjust the formula is as follows:

$$\omega_{ij}(t+1) = \omega_{ij}(t) + \eta(t,N)\left[x_{ip} - \omega_{ij}(t)\right] \tag{4}$$

In the formula (4), i is the number of iterations. ω_{ij} represents the weight of the j-th neuron during the i-th iteration. η is the learning rate.

Step 5: After multiple iterations, the positions of output layer nodes are determined. The similarity is determined based on the Euclidean distance between each component and the standard component. To avoid the occurrence of low probability events, the classification process is iterated 1000 times, and the Euclidean distance after each classification is recorded. Components with Euclidean distance less than 2 from the standard component and occurring more than 600 times are considered as the same type of signal containing noise.
Step 6: Wavelet soft threshold denoising is used to process the classified noise signal. On the basis of data feature mining, the signal adaptive noise reduction is realized.

2.2. Improved seagull optimization algorithm

2.2.1. Seagull optimization algorithm

Seagulls are social seabirds that migrate in groups as the seasons change [58]. SOA is a new swarm intelligent optimization algorithm which simulates the migration and foraging behavior of seagulls in nature. Migration behavior refers to the process in which seagulls move towards the optimal location from their current position, and during this stage, individual seagulls have flight independence. Foraging behavior refers to the predatory behavior of seagulls on prey on the water or land during flight, during which seagulls will spiral toward the prey to launch an attack [59]. The algorithm design of migration behavior affects the global search ability of the seagull algorithm, while the foraging behavior determines the local optimization ability of the seagull algorithm.

In the migration stage, attention should be paid to avoid the collision, and the position of seagulls keeps getting closer to the optimal position. The formula for calculating the new position C_s of seagulls to avoid collisions is as follows:

$$\begin{cases} C_s(t) = A \cdot P_s(t) \\ A = f_c(1 - t/Max_{iteration}) \end{cases} \tag{5}$$

In the formula (5), P_s represents the current optimal position. A is the motion behavior of a seagull in the search space. f_c is a function that controls the frequency of change of A, decreasing linearly from 2 to 0. $Max_{iteration}$ is the maximum number of iterations. The seagull will move to the optimal position, and the movement formula is shown as follows:

$$\begin{cases} M_s(t) = B \cdot (P_{bs}(t) - P_s(t)) \\ B = 2 \cdot A^2 \cdot r_d \\ D_s = |C_s + M_s| \end{cases} \tag{6}$$

$M_s(t)$ is the direction of the optimal position. $P_{bs}(t)$ is the current optimal position. $P_s(t)$ is the current position. B is a random number used to balance global and local search, and r_d ranges from 0 to 1. After obtaining the convergence direction of the seagull, the seagull will move along this direction, and D_s will be the seagull's new position.

Seagulls descend in a spiral pattern as they forage, their flight angle and radius constantly changing. The specific position of the seagull in three-dimensional space is:

$$\begin{cases} x = r \cdot \cos(\theta) \\ y = r \cdot \sin(\theta) \\ z = r \cdot \theta \\ r = u \cdot e^{\theta v} \end{cases} \tag{7}$$

In the formula, r is the spiral radius of seagull movement, θ is the angle value of the value range, u and v are spiral shape parameters. The formula of the seagull's update position is as follows:

$$P_s(t) = (D_s \cdot x \cdot y \cdot z) + P_{bs}(t) \tag{8}$$

2.2.2. ISOA based on cosine function, cauchy mutation and perturbation operator

To further improve the convergence speed of the SOA and enhance its search ability, this research proposes a hybrid strategy-improved Seagull Optimization Algorithm. By combining cosine function, Cauchy mutation and perturbation operator, the optimization algorithm is improved from the migration and position correction of seagulls [60].

In formula (5), A affects the migratory behavior of seagulls. It can be seen from the formula that A linearly decreases from f_c to 0. To improve the local search ability of seagulls in the early stage and the convergence speed in the middle stage, a cosine function is introduced to define A as:

$$A = f_c \cdot \cos\left(\pi \cdot \frac{t}{Max_{iteration}}\right) \Big/ 2 + 1 \tag{9}$$

Perturbation operators are added to the seagull position updating process to improve the ability of SOA to jump out of local optimal. When

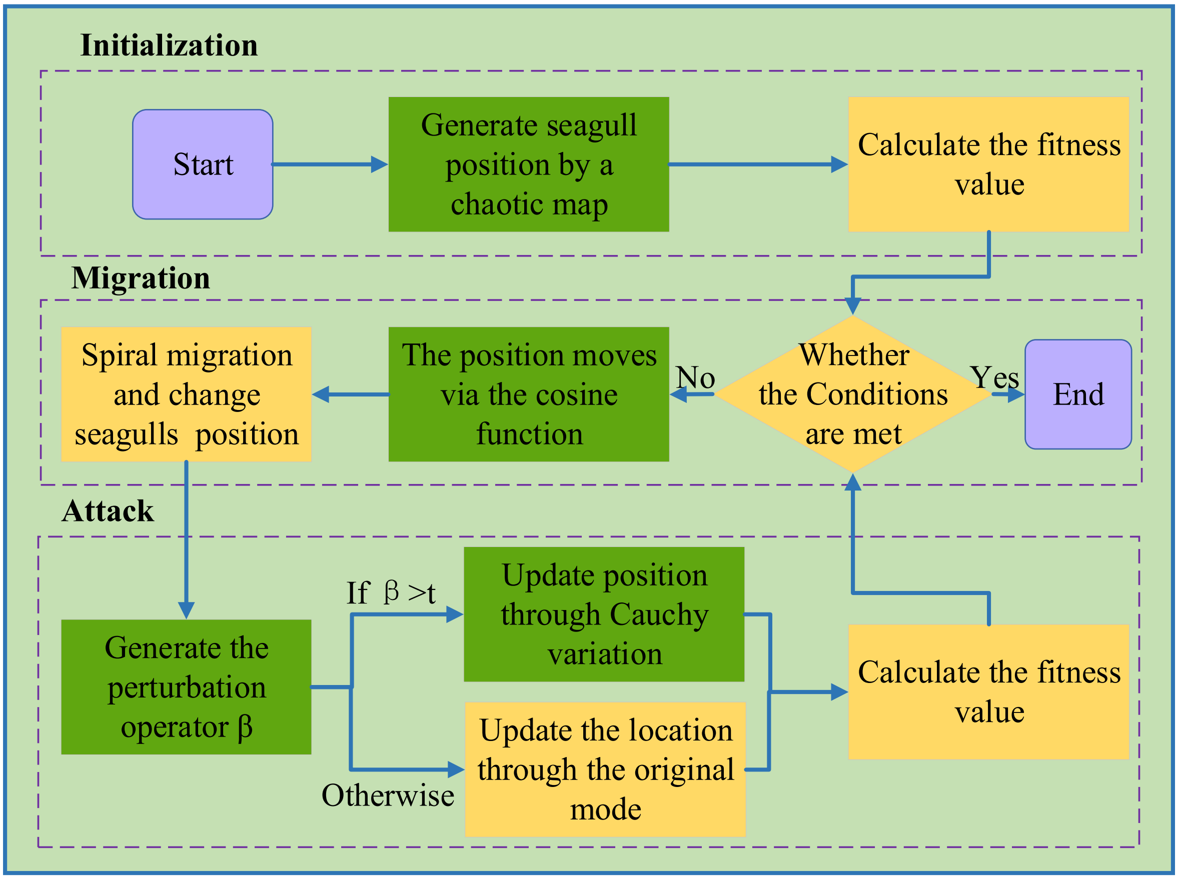


Fig. 2. Flow chart of ISOA based on hybrid strategy and chaotic mapping.

the perturbation operator satisfies $\beta < (1 - t/Max_{iteration})$, the position updating formula is formula (10); otherwise, Cauchy variation is added in the process of position updating, then:

$$P_s(t) = (D_s \cdot x \cdot y \cdot z) \cdot (1 + cauchy) + P_{bs}(t) \tag{10}$$

The Cauchy distribution is similar to the standard normal distribution, but has smaller values at the origin and is flatter at the tails, allowing for greater disturbance [61]. When combined with perturbation operators and Cauchy mutation, it can enhance the search capability of the SOA. The standard Cauchy distribution formula is shown in formula (11).

$$cauchy(x;0,1) = \frac{1}{\pi(1+x^2)} \tag{11}$$

2.2.3. ISOA based on hybrid chaotic mapping

Although SOA has good optimization capabilities, the random initialization process of the population increases the possibility of getting stuck in local optima. To enhance the diversity of the population, this work introduces the Logistics-Sine chaotic mapping [62] into the algorithm.

The logistics chaotic system has a simple structure and can generate complex chaotic characteristics. However, the Logistics chaotic range is limited, and when the input parameters are not within the range, the generated chaotic sequence becomes uneven [63]. Therefore, by integrating the one-dimensional Logistics chaotic system with the Sine chaotic system, the resulting compound chaotic system has complex chaotic dynamics, faster iteration speed, and is suitable for a large number of sequences [64]. Chaotic mapping optimization uses chaotic sequences to generate random numbers. It can effectively increase the coverage of the initial solution space, make the population closer to the optimal solution faster, and accelerate the convergence rate of the algorithm.

To better illustrate the improvement process of ISOA, this work combines Section 2.2 and draws the overall framework diagram of ISOA as shown in Fig. 2.

2.3. Long short-term memory network (LSTM)

Long Short-Term Memory (LSTM) is a deep learning-based Recurrent Neural Network (RNN) that inherits the advantages of RNN models. Compared to traditional RNNs, LSTM does not suffer from the vanishing gradient problem and can better store and access information. The basic form of an LSTM structure includes input gates, output gates, forget gates, and memory cell states [65]. The three gates can control the memory cell state [66]. The input gate determines the information that will be stored in the memory cell state, the output gate determines the information that will enter the next hidden state, and the forget gate determines whether to forget the information from the previous state [67]. The specific steps are as follows:

Step 1: The amnesia gate determines how many neurons U_{t-1} from the previous moment can be retained for the current moment U_t. The forgetting gate inputs include the output Y_{t-1} of the previous moment and the input L_t of the current time, and f_t is obtained by activating the function Sigmoid. The formula is as follows:

$$\begin{cases} f_t = \sigma\left(\omega_f \cdot [Y_{t-1}, L_t] + c_f\right) \\ \sigma(x) = \left(1 + e^{-x}\right)^{-1} \end{cases} \tag{12}$$

where, ω_f is the weight matrix of the forgetting gate, c_f is the bias vector, and σ is activation function Sigmoid. f_t is the probability that the last layer of neurons will be forgotten, and the value range is [0,1]. $f_t = 0$ means completely discarded, $f_t = 1$ means completely retained.

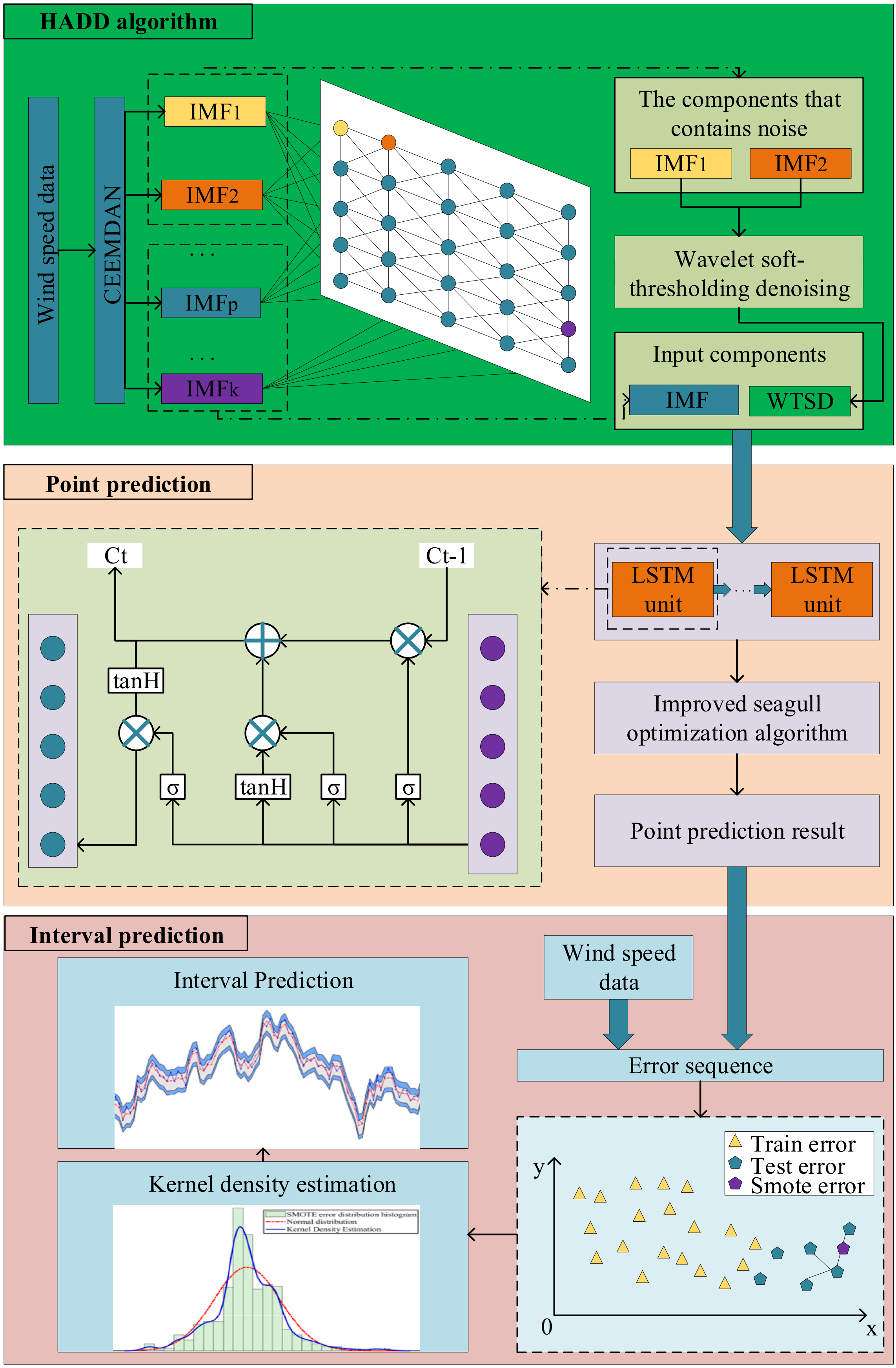


Fig. 3. The overall flow chart from point prediction to interval prediction.

Step 2: Input gates determine the new input in L_t that can be stored in neurons, which are mainly divided into p_t and q_t. The formula is as follows:

$$\begin{cases} p_t = \sigma\left(\omega_p \cdot [Y_{t-1}, L_t] + c_p\right) \\ q_t = \tanh\left(\omega_q \cdot [Y_{t-1}, L_t] + c_q\right) \\ U_t = U_{t-1} \cdot f_t + p_t q_t \end{cases} \tag{13}$$

In the formulas (13), ω_p and ω_q are the weight matrices of the input gates. c_p and c_q are the bias vectors. p_t and q_t indicate the proportion of load information that needs to be retained at the current time. The new neuron state information U_t consists of the neuron state retained at the previous time and p_t and q_t.

Step 3: The output gate is used to control the output state of neurons and transmit the state to the next neuron. The final output value can be obtained by the last layer of calculation.

$$\begin{cases} O_t = \sigma(\omega_O[Y_{t-1}, L_t] + c_O) \\ Y_t = O_t \cdot \tanh(U_t) \end{cases} \tag{14}$$

where, O_t is the output threshold. ω_o is the weight matrix of the output gate, c_o is the bias vector and Y_t is the output value at the current moment.

2.4. Interval estimation of wind speed based on SKDE method

Due to the limitations of actual wind speed prediction and engineering applications, only the wind speed prediction results for a certain period of time in the future can be obtained, resulting in a relatively small amount of error data. The simple replication of error samples in the Bootstrap method can not introduce new effective information, and may lead to overfitting of errors [68]. Therefore, the error processing method based on data augmentation technology has attracted the attention.

SMOTE algorithm is an improved solution of a random oversampling algorithm. It can generate new samples by linear interpolation between two existing samples, based on their relationship to expand the dataset [69]. Kernel density estimation (KDE) is a non-parametric estimation method for inferring the probability density function of an unknown variable [70]. This method does not require any assumptions about the distribution function and relies entirely on the distribution characteristics of the data sample itself. Therefore, based on the distribution characteristics of the error data, the KDE method can provide interval estimates for wind speed prediction [8]. The steps of wind speed interval estimation based on SKDE method are shown as follows:

Step 1: Construct imbalanced classes based on point prediction results.

According to the division of the original data set, the ratio of the number of sampling points between the training set and the test set is M/N (M > N). The wind speed point prediction model is obtained according to the training set. Then, the error sequence $e_1, e_2, \ldots, e_M$ of the training set and $e_1, e_2, \ldots, e_N$ of the test set are obtained according to the model.

Step 2: Oversampling of error based on SMOTE method.

Firstly, each e in the test set error is selected as the starting point for oversampling in turn. Then, k nearest neighbor samples is found. A random one is chosen from the nearby samples each time to generate new error samples using linear interpolation. This process is repeated n times.

$$e_{ij,new} = e_i + \eta \cdot \left(e_{ij} - e_i\right) \tag{15}$$

where, e_i is the i-th sample in the error sequence of the test set and e_{ij} is the j-th adjacent sample of e_i. η is the random number between [0,1], and $e_{ij,new}$ is the new error value obtained by interpolation.

Step 3: Interval estimation of wind speed.

The wind speed interval is established according to the expanded error data. The KDE method is selected to fit the error distribution. The formula for KDE is as follows:

$$\widehat{f}_m(x) = \frac{1}{n}\sum_{i=1}^{n} K_m(x - x_i) = \frac{1}{nm}\sum_{i=1}^{n} K_m\left(\frac{x - x_i}{m}\right) \tag{16}$$

where, $x_1, x_2, \ldots, x_n$ is a random variable distributed independently and identically according to the probability density function $f(x)$. m is a smoothing parameter, and $K(\bullet)$ is a kernel function. In this research, the Gaussian kernel function is selected as the kernel function, and it is shown in the formula (17).

$$K(x, \mathrm{m}) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x - x_i)^2}{2m^2}\right) \tag{17}$$

Fit the error sequence and calculate the probability density function $\widehat{f}_m(x)$. According to the confidence level, different sub-sites were obtained to construct the wind speed prediction interval.

3. Establishment of prediction model

To address the intermittency and variability of wind speeds, this work proposes a novel hybrid wind speed prediction system that combines data decomposition, algorithm optimization, and deep learning models. Using data augmentation techniques to generate wind speed prediction intervals, the fluctuation range of wind speed can be effectively quantified. First, the original data are decomposed by CEEMDAN, and then the SOM is applied to classify the components, and combined with WST denoising to remove the residual white noise to improve the prediction robustness. Secondly, the LSTM model is adopted to predict the wind speed using the denoised component signals. The ISOA is employed to enhance the search and convergence ability of the algorithm. Finally, the error data is processed using data augmentation techniques. KDE method is used to process the error sequence, and wind speed interval prediction is established. The flow chart of wind speed prediction model established is shown in Fig. 3.

The specific steps of the hybrid wind speed prediction system established in this research are as follows:

Step 1: Wind speed data were selected from the datasets collected from the Eman wind farm in China and the Sotavento wind farm in Spain. The proposed HADD method was used to decompose and filter the data, generating component sub-sequences through iterations.
Step 2: Chaotic mapping and Cauchy mutation algorithm are used to improve the global search capability and convergence speed of SOA. The parameters of deep learning model LSTM are optimized by ISOA. Using the optimized LSTM model forecast component, the wind speed point prediction results are obtained.
Step 3: Four kinds of prediction evaluation indicators are used to estimate the point prediction results of the proposed model, which are the mean absolute error (MAE), root mean square error (RMSE), mean absolute percentage error (MAPE), and coefficient of determination (R^2). The prediction evaluation metrics formula is as follows:

Table 1
Statistical index of wind speed distribution.

Wind farm	Max (m/s)	Min (m/s)	Mean (m/s)	Std. (m/s)	Lower quartile(m/s)	Upper quartile (m/s)
Sotavento	12.26	0.02	6.58	2.36	5.02	8.19
Eman	10.05	0	4.07	2	2.6	5.25

$$\begin{cases} MAE = \dfrac{1}{m}\sum_{i=1}^{m}(y_i - \widehat{y}_i) \\ RMSE = \left(\dfrac{1}{m}\sum_{i=1}^{m}(y_i - \widehat{y}_i)^2\right)^{\frac{1}{2}} \\ MAPE = \dfrac{1}{m}\sum_{i=1}^{m}\dfrac{|y_i - \widehat{y}_i|}{y_i} \\ R^2 = 1 - \dfrac{\sum_{i=1}^{m}(y_i - \widehat{y}_i)^2}{\sum_{i=1}^{m}(y_i - \overline{y})^2} \end{cases} \tag{18}$$

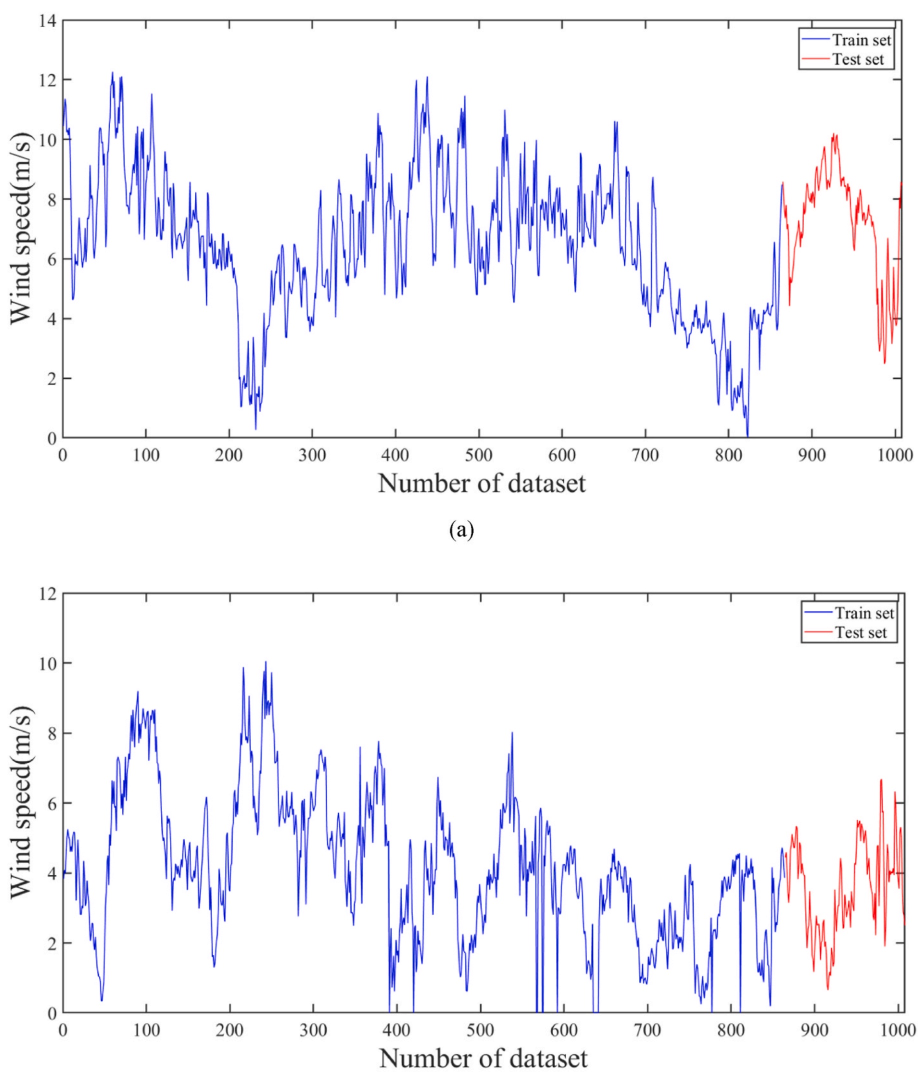
where y_i represents the true series data, $\widehat{y}_i$ represents the predicted data, $\overline{y}$ is the series mean, and m is the number of series samples.

Step 4: Based on the point prediction results, a data enhancement algorithm is introduced. The prediction interval of wind speed is obtained by oversampling and expanding error data combined with kernel density estimation algorithm.

Step 5: The interval prediction effect is evaluated by interval evaluation index. PICP represents the coverage probability of the prediction interval. PINAW is the normalized average width of the prediction interval, which represents the width of the interval at different confidence levels. To better evaluate the interval prediction results, taking into account the accuracy and reliability of the prediction, the coverage standard width CWC and the average interval score AIS are introduced by considering both interval coverage and interval width. Under the same confidence level, the smaller the CWC value represents the better interval prediction; the larger the AIS value represents the better interval prediction.

$$PICP = \frac{\sum_{i=1}^{m} A_i}{m} \tag{19}$$

$$PINAW = \frac{\sum_{i=1}^{m} \xi_i}{m} \tag{20}$$



(a)

(b)

Fig. 4. Data sets of Sotavento and Eman farm, (a) Wind speed of Sotavento farm, (b) Wind speed of Eman farm.

Table 2
Distribution of the components.

	Sotavento	Eman		Sotavento	Eman
IMF1	1000	1000	**IMF6**	553	2
IMF2	776	909	**IMF7**	126	5
IMF3	522	870	**IMF8**	17	23
IMF4	678	385	**IMF9**	3	45
IMF5	748	5	**IMF10**	0	34

$$CWC = PINAW\left(1 + \lambda e^{-\eta(PICP-\alpha)}\right) \tag{21}$$

$$AIS = \frac{\sum_{i=1}^{m} S_i}{m} \tag{22}$$

$$S_i = \begin{cases} -2\alpha\xi_i - 4(L_i - \widehat{y}_i) & \widehat{y}_i < L_i \\ -2\alpha\xi_i & L_i \le \widehat{y}_i \le U_i \\ -2\alpha\xi_i - 4(\widehat{y}_i - U_i) & \widehat{y}_i > U_i \end{cases} \tag{23}$$

In the formula (23), $A_i = 1$ when the true value is within the prediction interval, otherwise, $A_i = 0$. ξ_i is the width of the interval. The value of λ is related to the confidence level. When the PICP is greater than the confidence level, $\lambda = 1$, otherwise, $\lambda = 0$. S_i is the interval fraction of the prediction interval. L_i and U_i are the lower and upper bounds of the prediction interval, respectively.

4. Wind speed point prediction results and error analysis

4.1. Data introduction

This work selects wind speed data from Sotavento wind farm in Spain and Eman wind farm in China as data sets(see Table 1). The interval between both wind farms' data is 10 min. Fig. 4(a) displays wind speed data from Sotavento wind farm, where the wind speeds are mainly distributed in the range of [5.015, 8.19]. Fig. 4(b) displays wind speed data from Eman wind farm, where the wind speeds are mainly distributed in the range of [2.6, 5.245]. It can be observed that both wind farms have significant wind speed fluctuations, with different peak values at different times, and there are large differences in peak values before and after, indicating strong randomness. This research selects one week's wind speed data from both wind farms, and 144 sampling points on the last day are used as the prediction set, represented by a red curve, while the training set is represented by a blue curve.

4.2. Hybrid adaptive decomposition denoising

The original wind speed of wind farm data is processed by HADD. First, the CEEMDAN method was used to decompose the wind speed and obtain the corresponding IMF sequence. High-frequency components may retain white noise, thus IMF components containing noise need to be identified and denoised.

Input the components obtained by decomposition into the SOM neural network model. Set IMF1 component as the standard component sample and normalize the input components. Calculate the Euclidean distance between the remaining components and the IMF1 component, and determine the similarity of the components by calculating the distance. To reduce experimental randomness, the classification is repeated 1000 times. Record the IMF components with Euclidean distance to the standard component less than 2 units, and consider the components with repetition frequency greater than 600 to contain noise. The cumulative frequency of components is shown in Table 2, which shows that some components have repetition frequency greater than 600. This indicates that these components are very similar to the IMF1 component and have noise residues as judged by the SOM neural network. In the Sotavento wind farm, IMF1, IMF2, IMF4, and IMF5 are considered as noise signal components. In the wind speed data components of the Eman wind farm, IMF1, IMF2, and IMF3 are considered as signal components with noise. The signal components containing noise were subjected to wavelet soft-threshold denoising separately to reduce the residual noise effect. HADD not only implements data decomposition, reducing the complexity of prediction, but also achieves adaptive signal denoising, further improving prediction accuracy. The decomposed components of the original wind speed data obtained through mixed adaptive decomposition denoising are shown in Figs. 5 and 6.

4.3. Point prediction model validation

In this research, the feasibility and validity of the prediction model are verified using a variable control method. First, the validity of the selected LSTM model is verified by comparing the model with the BP model, Elman model, LSSVM model, and RBF model. The specific prediction errors are shown in Fig. 7. The model parameter settings here are

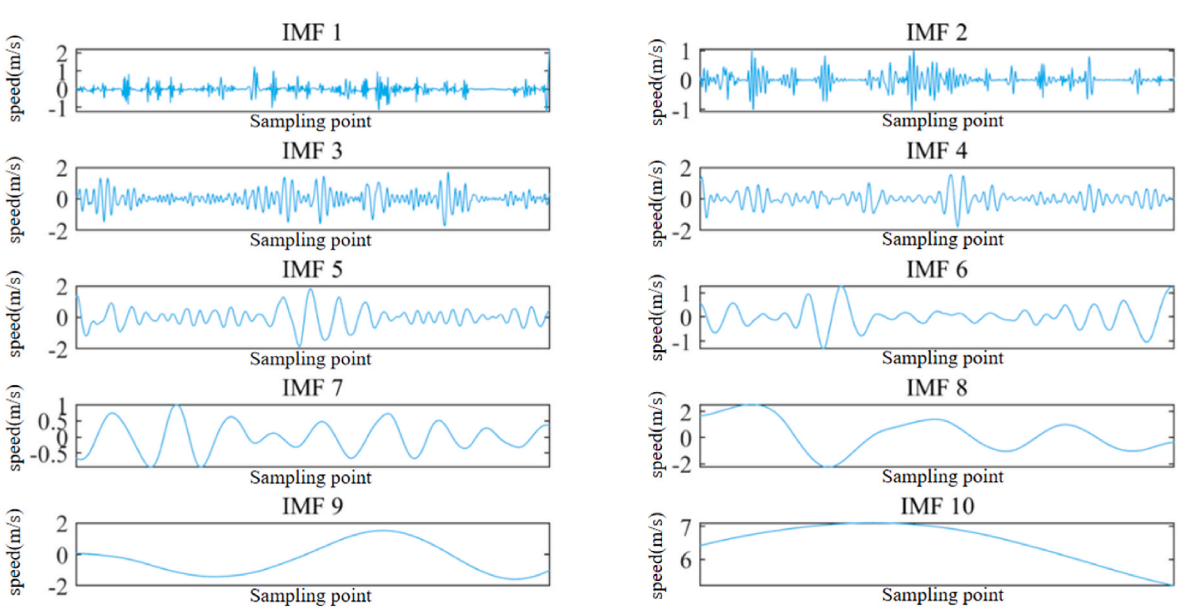


Fig. 5. HADD components of Sotavento.

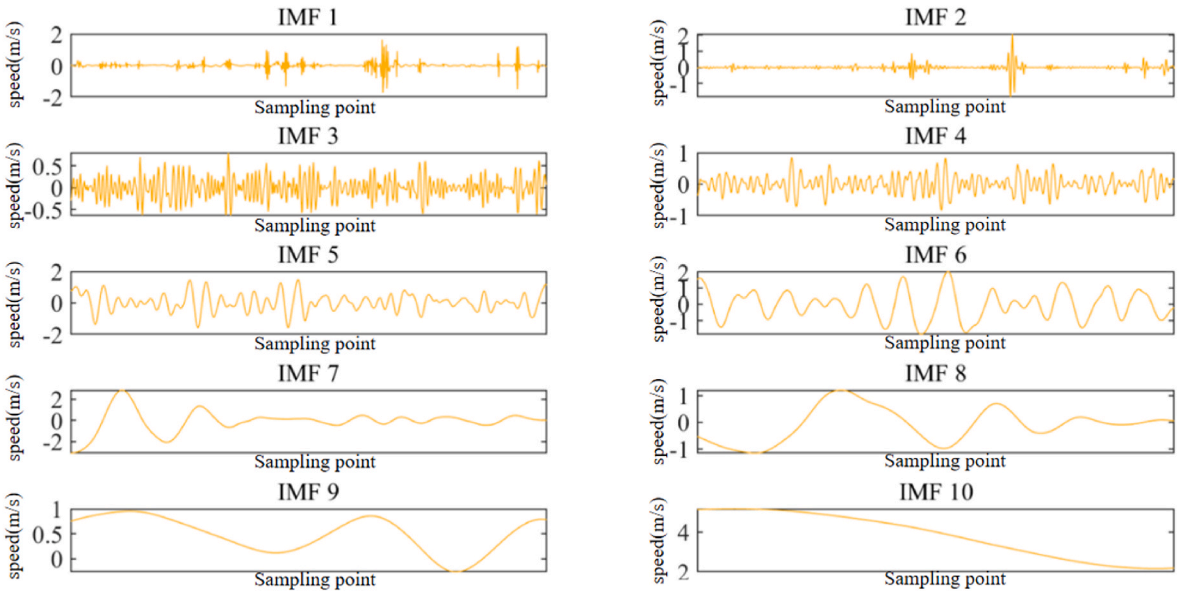


Fig. 6. HADD components of Eman.

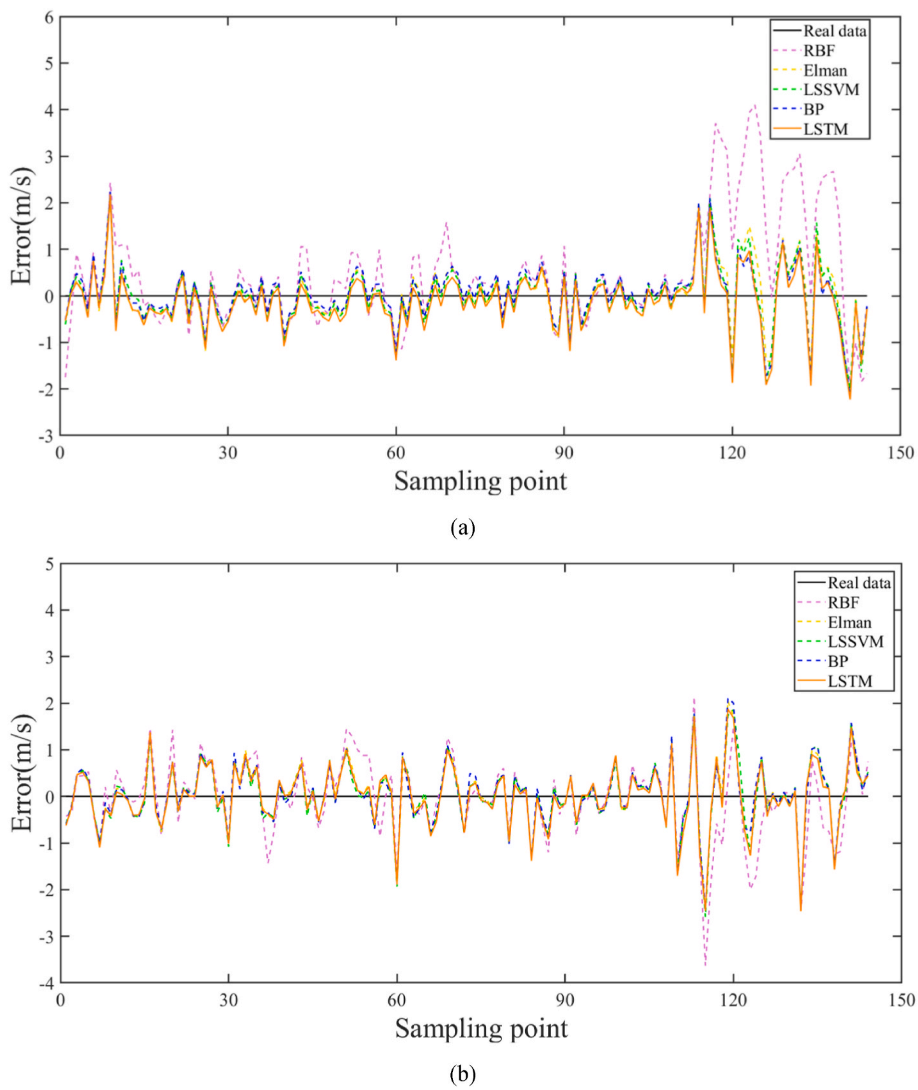


Fig. 7. Prediction error of single models, (a) Prediction error of Sotavento, (b) Prediction error of Eman.

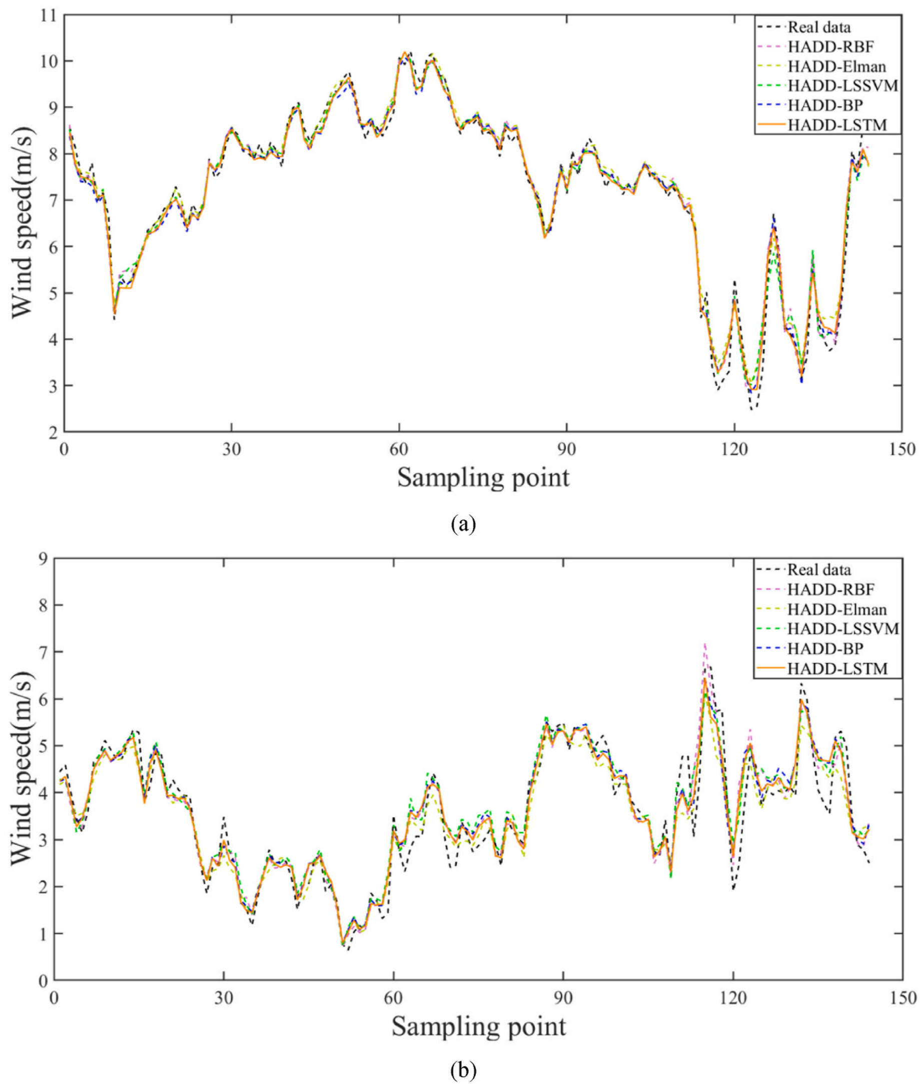


Fig. 8. Prediction results after using HADD, (a) Prediction results of Sotavento, (b) Prediction results of Eman.

based on extensive literature and experimental testing.

From Fig. 7, it can be seen that the purple dashed line represents the prediction error of the RBF model, the yellow curve represents the prediction error of the Elman model. While the blue curve represents the prediction error of the BP model, and the orange solid line represents the prediction error of the LSTM model. Although the prediction effect of a single model is poor and has obvious IMFtime lag. A single model can achieve preliminary prediction of wind speed, and the LSTM and BP models have better prediction performance. In the Eman wind farm, the data fluctuates significantly and the LSTM demonstrates better prediction performance than the BP model. It shows the practical significance of choosing deep learning model as prediction model.

To verify the impact of mixed adaptive decomposition denoising method on prediction accuracy, the original data of two wind farms are first decomposed by the mixed adaptive decomposition denoising method before prediction. Some components had complex fluctuations, but some reflected the main trend of the original wind speed. The partial components of Sotavento wind farm decomposition fluctuate relatively stable, the amplitude is small, but the signal frequency domain is high and the bandwidth is long. In combination with Tables 2 and it can be seen that residual white noise exists in IMF1, IMF2, IMF4 and IMF5 of Sotavento wind farm identified by HADD method, and Fig. 5 is obtained after de-noising. The processed component sequences are input into the model for prediction, and the prediction results are shown in Fig. 8(a). Similarly, the HADD-denoised Eman wind farm data components were input into different models for prediction, and the obtained prediction curves were displayed in Fig. 8(b).

From Fig. 8, it can be observed that the model prediction accuracy is significantly improved after being processed by the HADD method. All of the prediction curves effectively reflect the changing trend of the original wind speed data curve, and the prediction accuracy is greatly improved for wind speed abrupt changes compared to the single model. Combining Figs. 8 and 9, and Table 3, it can be seen that the prediction error is greatly improved after the HADD decomposition. Taking Sotavento wind farm as an example, after adding HADD, the prediction results demonstrate a significant decrease in MAE and RMSE. The RMSE value of HADD-BP decreased by 61.69% compared to the BP model, and the MAE value of HADD-LSTM decreased by 61.08% compared to the LSTM model, with a decrease of 59.26% in MAPE and an increase of 11.5% in regression coefficients. In Eman wind farm, the prediction set fluctuates more obviously, resulting in larger errors in some peak-to-valley prediction values. However, the prediction results after adding HADD show a significant improvement in error indicators. After decomposition and denoising, the prediction results improved by over 30% in both MAE and RMSE for the RBF model. The RMSE value decreased from 0.8185 to 0.4238 and the R^2 value increased from 0.6074 to 0.8948. Compared with the LSTM model, the RMSE value of HADD-LSTM decreased by 42.77% and the MAPE value decreased by

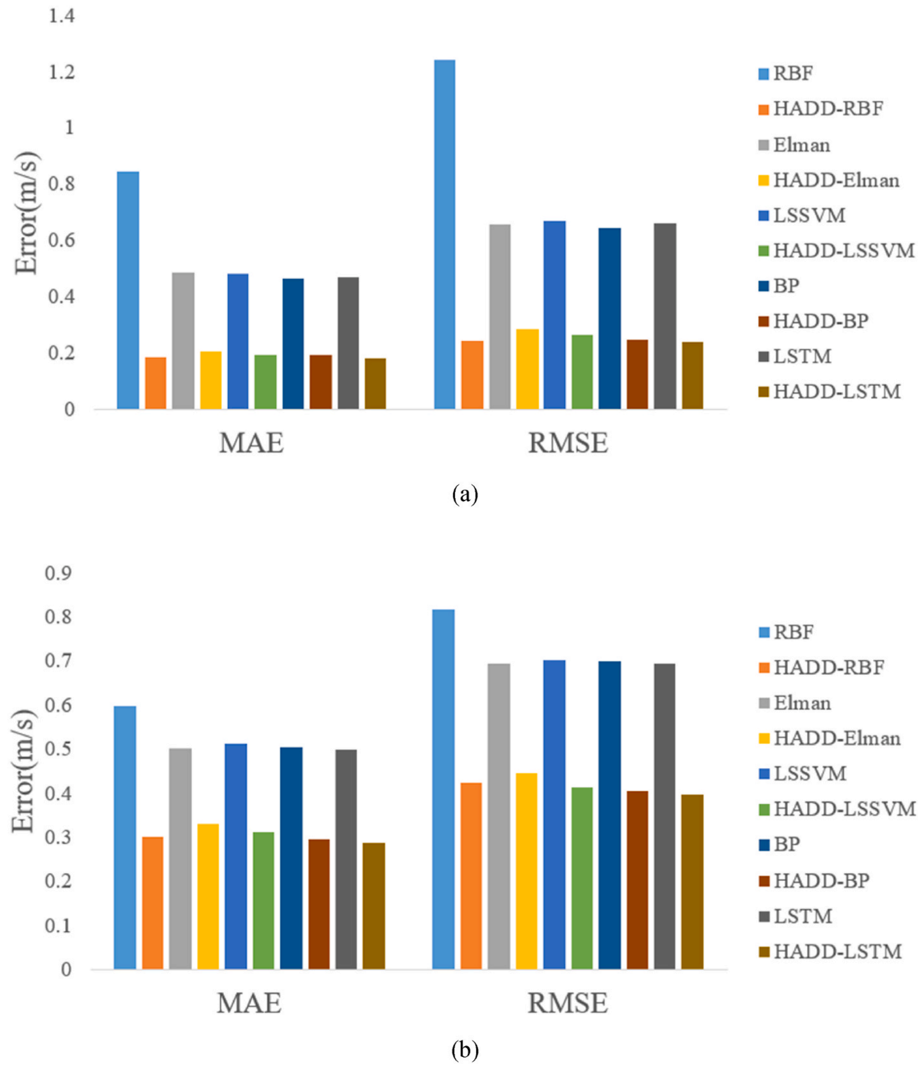


Fig. 9. Prediction error after using HADD, (a) Prediction error of Sotavento, (b) Prediction error of Eman.

Table 3
Prediction evaluation indicators of models.

	Sotavento				Eman			
	MAE (m/s)	RMSE (m/s)	MAPE	R^2	MAE (m/s)	RMSE (m/s)	MAPE	R^2
RBF	0.8448	1.2436	17.55%	0.5636	0.5973	0.8185	21.37%	0.6074
Elman	0.4869	0.6586	8.58%	0.8701	0.5014	0.6959	16.99%	0.7162
LSSVM	0.4825	0.672	8.34%	0.8647	0.5116	0.7035	16.93%	0.7099
BP	0.4649	0.6468	7.84%	0.8747	0.5039	0.6999	16.94%	0.7131
LSTM	0.4687	0.6617	7.8%	0.8693	0.4999	0.6935	16.69%	0.718
HADD-RBF	0.1869	0.2465	3.33%	0.9818	0.3016	0.4238	9.68%	0.8948
HADD-Elman	0.2093	0.2875	3.94%	0.9753	0.3315	0.4464	10.22%	0.8834
HADD-LSSVM	0.196	0.2644	3.55%	0.9791	0.3121	0.4149	10.44%	0.8998
HADD-BP	0.194	0.2478	3.31%	0.9816	0.2954	0.4068	9.77%	0.9033
HADD-LSTM	0.1824	0.2414	3.19%	0.9825	0.2888	0.3969	9.42%	0.9077
SOA-LSTM	0.4608	0.6504	7.77%	0.8733	0.4867	0.6839	15.65%	0.7273
ISOA-LSTM	0.4449	0.6238	7.2%	0.8836	0.4205	0.5538	14.07%	0.8203
HADD-SOA-LSTM	0.181	0.2343	3.13%	0.9836	0.2757	0.3693	9.42%	0.9204
HADD-ISOA-LSTM	0.1682	0.2199	2.87%	0.9855	0.2402	0.3052	8.01%	0.9454

43.56%. Based on the error evaluation indicators, further optimization and improvement based on HADD-LSTM can have a positive effect on improving wind speed prediction accuracy.

4.4. *Verify the effectiveness of the ISOA*

The SOA is an intelligent swarm optimization search algorithm that exhibits excellent performance in model parameter optimization. To further improve the search ability of the SOA, accelerate the convergence speed, and increase the population diversity of the algorithm, this

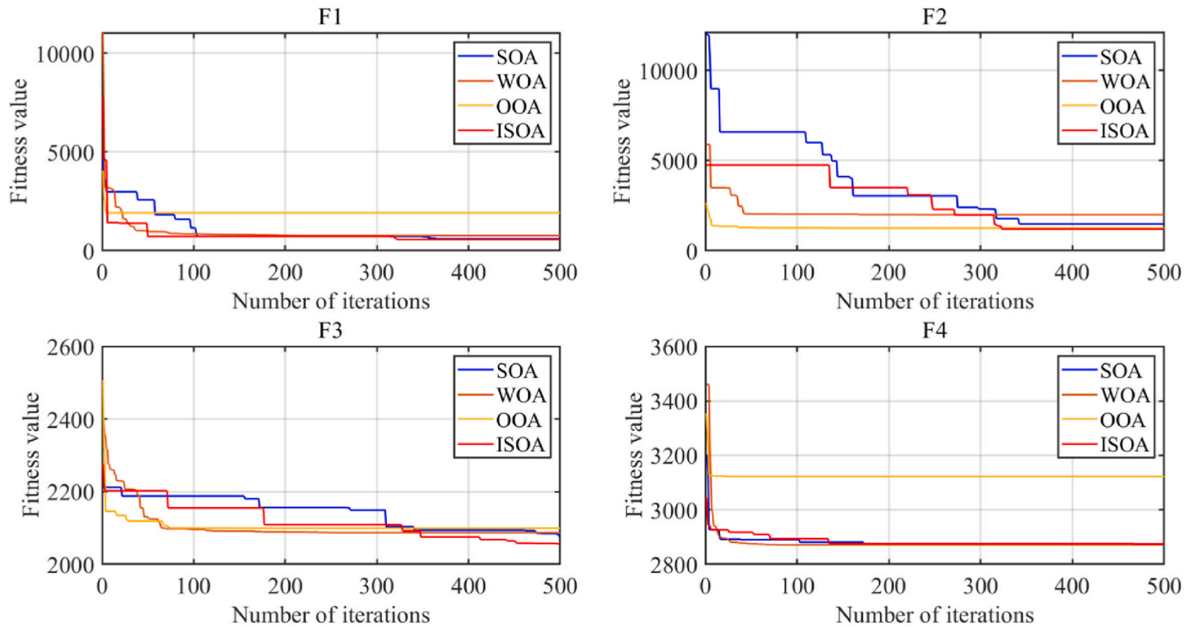


Fig. 10. Fitness curve of CEC2022 test functions.

work proposes an ISOA that uses chaotic mapping to increase the initial population diversity. The cosine function, Cauchy mutation, and perturbation operator are introduced to enhance the global search ability.

The initialization process of the SOA is improved by logistics-Sine hybrid chaotic mapping system. Using chaotic sequences to generate random numbers, the random initialization in the solution space is more evenly distributed, making the population reach the relatively optimal solution more quickly.

The cosine function is used to transform linear transformations into non-linear transformations to change the movement behavior of the seagull. Cauchy mutation and perturbation operators are added. When the algorithm iteration falls into a local optimum, these operators can increase the possibility of the algorithm jumping out of the local optimum and improve the convergence ability of the algorithm.

To illustrate the superiority of the improved algorithm, this work introduces two optimization algorithms with superior performance, WOA and OOA, which were proposed in 2016 and 2023 respectively. The CEC2022 test function is used to compare traditional SOA and ISAO, and the effectiveness of ISOA in improving search efficiency and avoiding local optimality is verified. Compared with WOA and OOA algorithms, ISOA's performance is superior. The red curve shows the change in the fitness value of the ISOA function.

From Fig. 10, it can be observed that compared to SOA, in the early iterations, ISOA with the inclusion of a hybrid chaotic mapping exhibits lower fitness values and a faster declining trend. In the later iterations, ISOA considering the Cauchy mutation and perturbation operator demonstrates a better ability to escape local optima and tends towards the global optimum, showcasing stronger convergence capability. Compared to WOA and OOA, ISOA does not show a significant advantage in the early iterations. However, in the later iterations, ISOA displays a higher ability to escape local optima. This advantage is more pronounced in the relatively complex F3 and F4. The performance of ISOA indicates that the inclusion of a hybrid improvement strategy leads to a notable enhancement in both convergence capability and the ability to escape local optima.

Using ISOA to optimize the parameters of the LSTM model and combining with the HADD algorithm, the predicted results are shown in Fig. 11. From Fig. 11 and Tables 3 and it can be seen that for Sotavento wind farm as an example, after adding optimization algorithm, the predictive performance of both SOA-LSTM and ISOA-LSTM models improved compared to the LSTM model, with RMSE values decreased by 1.71% and 5.73%, respectively. Compared to the SOA-LSTM model, the ISOA-LSTM model reduced the predictive MAE by 0.0159 m/s, or 3.45%. The MAPE value of the HADD-ISOA-LSTM model was 2.87%, which decreased by 8.31% compared to the HADD-SOA-LSTM model. In the Eman wind farm, the RMSE value of ISOA-LSTM decreased by 19.02% compared to SOA-LSTM. The MAPE value of HADD-ISOA-LSTM was 8.01%, which is a 14.97% decrease compared to HADD-SOA-LSTM. In the prediction results of the two wind farms, the HADD-ISOA-LSTM model incorporating the HADD algorithm and ISOA achieved an R^2 value of over 94%, indicating good fitting and high prediction accuracy.

Finally, this research compared and calculated the errors of single models and combined prediction models, and presented the error indicators in Table 3. The proposed HADD algorithm and ISOA have improved the accuracy of wind speed prediction, indicating that the improvement strategy proposed has a positive effect on improving wind speed prediction.

5. Wind speed interval prediction

The innovation and optimization of point prediction models for wind speed can improve the accuracy of prediction to a certain extent. However, it is difficult to evaluate the fluctuation of wind speed based on point predictions alone. To enhance the reliability of wind speed prediction, this work proposes a wind speed interval prediction method based on data augmentation techniques and kernel density estimation.

5.1. Wind speed range forecast based on SMOTE and KDE

Firstly, the error sequence of training set and test set is obtained using the wind speed point prediction model proposed and wind farm data. Then, use SMOTE to build most classes, the train-set error sequence, and a few classes, the test-set error sequence.

In the test set error, new error samples were generated based on neighboring samples. The frequency histogram is drawn based on the empirical cumulative distribution function. The horizontal axis of the graph represents the value of the prediction error from small to large, and the vertical axis represents the corresponding empirical distribution function value. Fig. 12 shows the original error distribution of the test

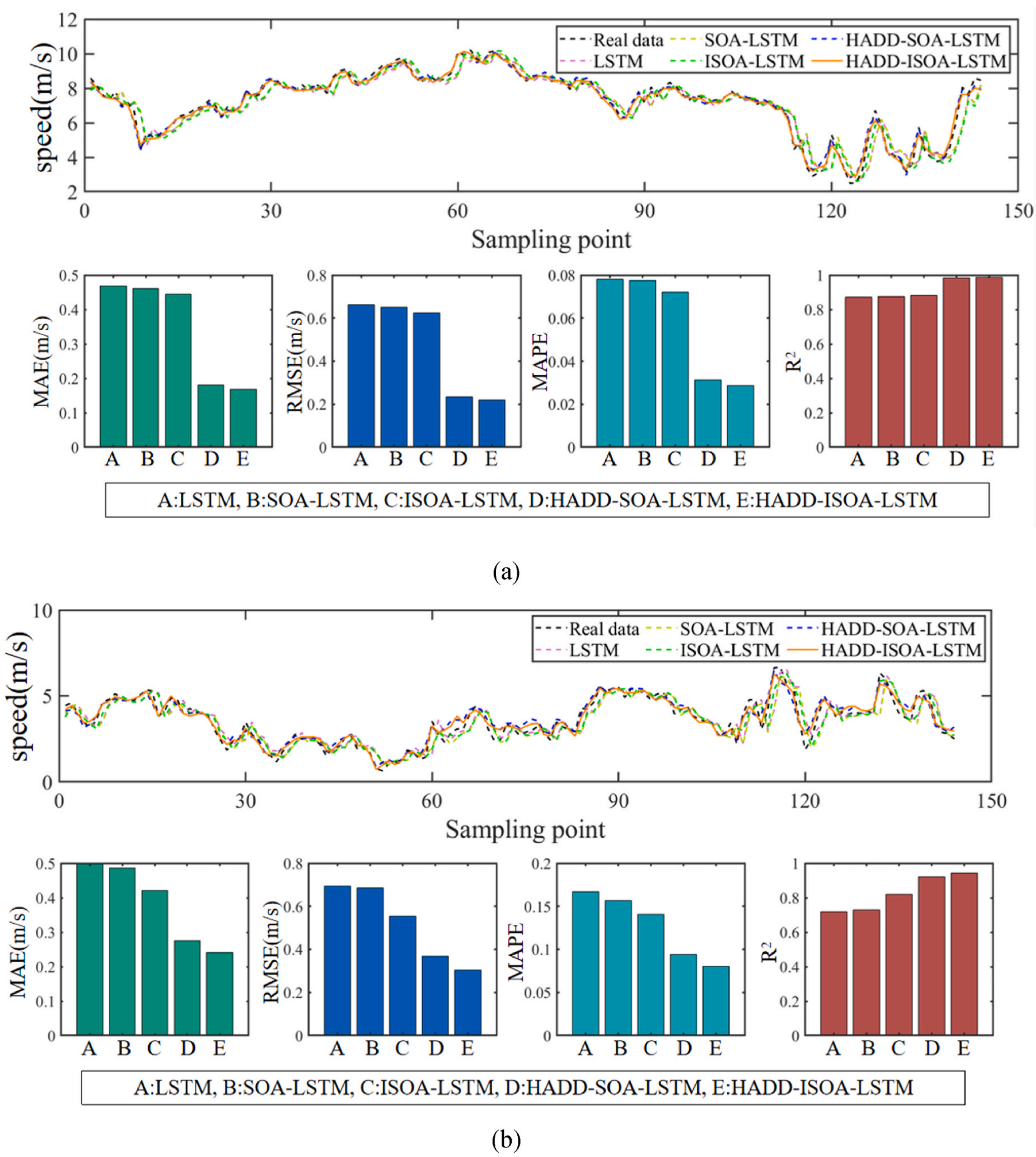


Fig. 11. Prediction results after using ISOA, (a) Prediction results of Sotavento, (b) Prediction results of Eman.

set for Sotavento and Eman wind farms and the error distribution obtained after SMOTE data augmentation. After augmentation, the number of error samples increased fourfold, but the distribution remained similar. Then, the probability density function of the error sequence was obtained using KDE. The distribution curve estimated by kernel density estimation is shown as the yellow line in Fig. 12(a) and (b), and it can be observed that KDE is effective in fitting the error distribution.

Different wind speed prediction intervals were generated based on different confidence levels. The results are shown in Fig. 13, indicating that the wind speed intervals can effectively provide the fluctuation range of the wind speed, and most of the actual data can be covered by the intervals. To better compare the interval prediction performance, this study also used different comparison models and error evaluation indicators for validation.

5.2. Verification of wind speed interval prediction effect

From Fig. 12, it can be seen that fitting two groups of errors with normal distribution has poor results and the Kolmogorov-Smirnov (KS) test result for the distribution of the two error sequences is 1, so the normal distribution method is not used for fitting. Using the SMOTE method to process the error data can avoid overfitting that may be caused by the Bootstrap method, generate effective new error data, and increase data volume without changing the data structure. Intuitively, the SMOTE-generated error sequence is similar in distribution to the original error sequence, and the fit of the normal distribution is poor. Therefore, this work uses KDE without prior knowledge of the sequence distribution characteristics and uses Gaussian function as the kernel function. The yellow curve representing the kernel density probability distribution can fit the error distribution well.

This research uses the quantile interval method to establish wind

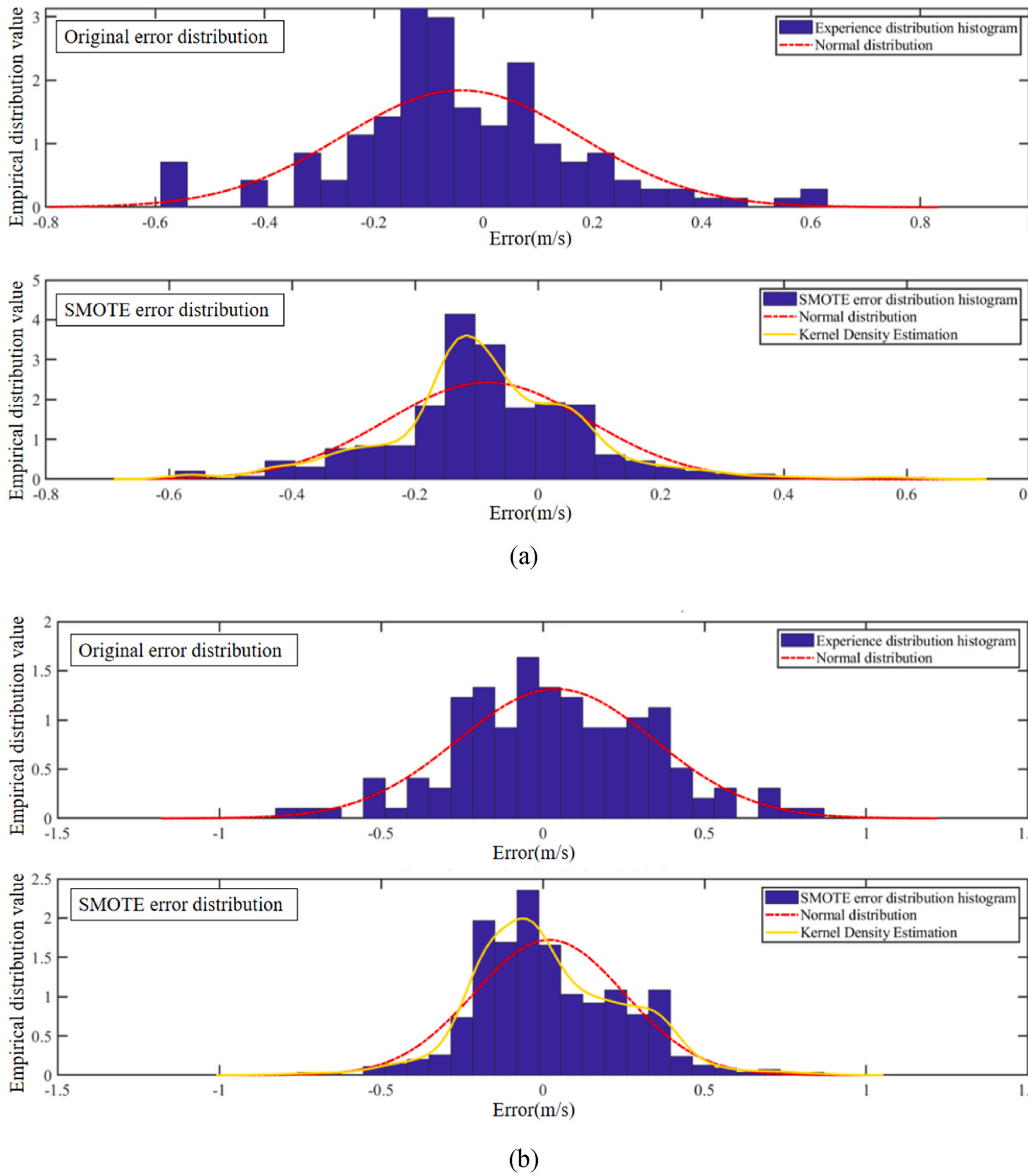


Fig. 12. Original error distribution and SMOTE error distribution, (a) Error distribution of Sotavento, (b) Error distribution of Eman.

speed intervals as a comparison model, and uses the CWC and AIS as error evaluation indicators for interval prediction. The results are shown in the Table 4.

Contrast the range given based on SMOTE and KDE with the quartile range. It can be seen that, due to strong wind speed fluctuation and high randomness, the prediction effect of wind speed interval is poor when the wind speed changes greatly and the peak-peak value and peak-valley part. However, under the 98% confidence level, the interval coverage of both wind farms can reach more than 93%, and the average interval width is small. Under the same confidence level, the average width of wind speed interval proposed is smaller than the quartile interval. To comprehensively consider the coverage rate and interval width of interval prediction, in terms of CWC, the proposed interval prediction model in this research has a smaller CWC value compared to the interval prediction results of two wind farms. This means that on the basis of comprehensively considering coverage rate and average interval width, the proposed model has better predictive performance. Similarly, according to AIS values, at the same confidence level, the AIS value of the proposed interval prediction model is larger than that of the quantile interval prediction, which also proves that the proposed interval prediction model can balance accuracy and reliability, effectively quantify wind speed fluctuations.

6. Conclusion

Accurate wind speed prediction is the basis of the development of wind power industry, and it is the necessary technical guarantee to reduce power generation cost and avoid resource waste. To improve the safety and reliability of the wind power forecasting system effectively, this study proposes a hybrid prediction system that combines data

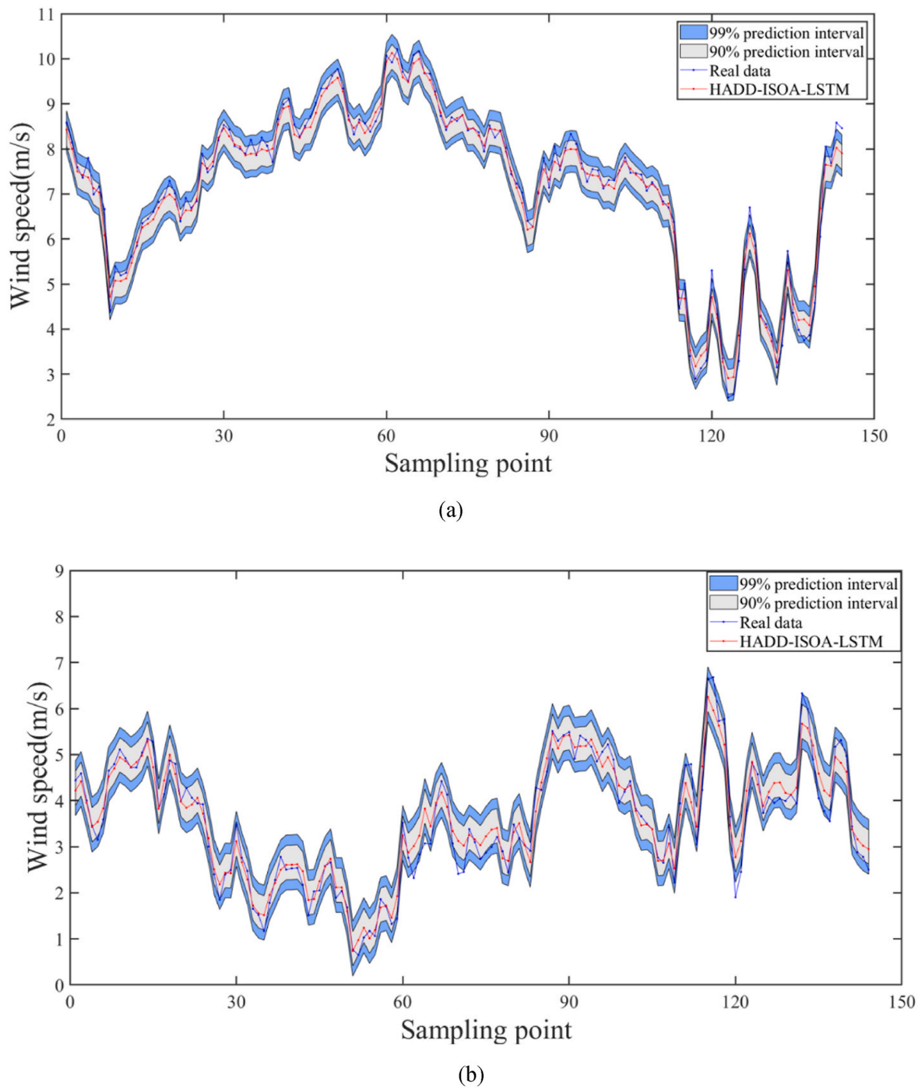


Fig. 13. Wind speed prediction intervals under different confidence levels, (a) Prediction intervals of Sotavento, (b) Prediction intervals of Eman.

Table 4
Evaluation indicators for interval prediction.

Data set	Confidence	Interval prediction	CWC	AIS
Sotavento	0.9	Quantile	1.4602	−1.3622
		SMOTE-Kernel	1.1894	−1.1286
	0.98	Quantile	2.3324	−2.2867
		SMOTE-Kernel	1.8996	−1.8451
Eman	0.9	Quantile	1.9905	−1.8375
		SMOTE-Kernel	1.6013	−1.491
	0.98	Quantile	2.9548	−2.9017
		SMOTE-Kernel	2.4405	−2.3686

processing, model optimization, deep learning, and interval prediction. Firstly, the HADD algorithm is used to adaptively decompose and denoise the original wind speed data, improving data quality and reducing prediction complexity. Secondly, the search capability of SOA is enhanced using methods such as chaotic mapping, Cauchy mutation, and perturbation operator, and the parameters of the LSTM are searched and optimized using ISOA. Finally, wind speed interval prediction is constructed by combining data augmentation techniques and the KDE method. The accuracy and reliability of the system are verified using historical data from the Sotavento wind farm in Spain and the Eman wind farm in China. The main conclusions are as follows:

(1) By proposing the HADD method, data decomposition can be adaptively performed to identify residual noise sequences and denoise noise components. This effectively improves data quality and avoids the occurrence of over-decomposition phenomena.
(2) The ISOA algorithm obtained through the hybrid improvement strategy has stronger global search ability and faster convergence speed. The ISOA is used to optimize the parameters of the LSTM model, which greatly improves the accuracy and stability of the prediction model.
(3) Using the SMOTE data augmentation algorithm to process error data can avoid overfitting problems. Combined with KDE method, interval estimation of wind speed sequences is achieved. The interval prediction model proposed takes into account both the accuracy and reliability of the prediction, making it an effective interval prediction method.
(4) The point prediction model established in this work can accurately predict the change of wind speed in the next period of time, provide technical support for power system dispatching and improve the safety. The interval prediction model established in this research takes into account the accuracy and reliability of wind power fluctuations, and can improve the proportion of clean energy.

The hybrid prediction model proposed in this study can be applied not only to wind farms, but also to photovoltaic power plants. The research improves the existing algorithm to some extent, but it still needs further improvement. This research assumes that the future wind speed is the same as when the model is train, and the possibility of sudden changes is excluded, so the prediction has limitations. In the future, it is necessary to consider the possibility of sudden change of wind speed and analyze the relevant influencing factors combined with statistical methods. Interval prediction needs to explore new prediction models to further expand the coverage range of wind speed intervals on the premise of ensuring accuracy.

CRediT authorship contribution statement

Yagang Zhang: All authors participated in the research discussion and provided their views on the results and writing. The division of labor is as follows. **Xue Kong:** Conceptualization, Methodology, Supervision, and, Writing – original draft, preparation. **Jingchao Wang:** Visualization, Software, and, Methodology. **Hui Wang:** Formal analysis, Investigation. **Xiaodan Cheng:** Data Curation, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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